Work done in collaboration with:

Maksim Zhdanov **Erik Bekkers** **Patrick Forré**

To appear at ICLR 2025 Workshop on Machine Learning Multiscale Processes

AdS GNN: PascalVOC

- We also tested it on a **segmentation task, PascalVOC-SP** from Long Range Graph Benchmark (Dwivedi et. al 2022)

- Given a point cloud of pixels as in right-hand panel, has to predict “what each pixel is” out of 20 classes (e.g. “sailboat”, “background”, etc.)

- Genuinely hard. Our model is again comparable to existing ones (e.g. the idea of the “shape” of an object is conformally invariant?)

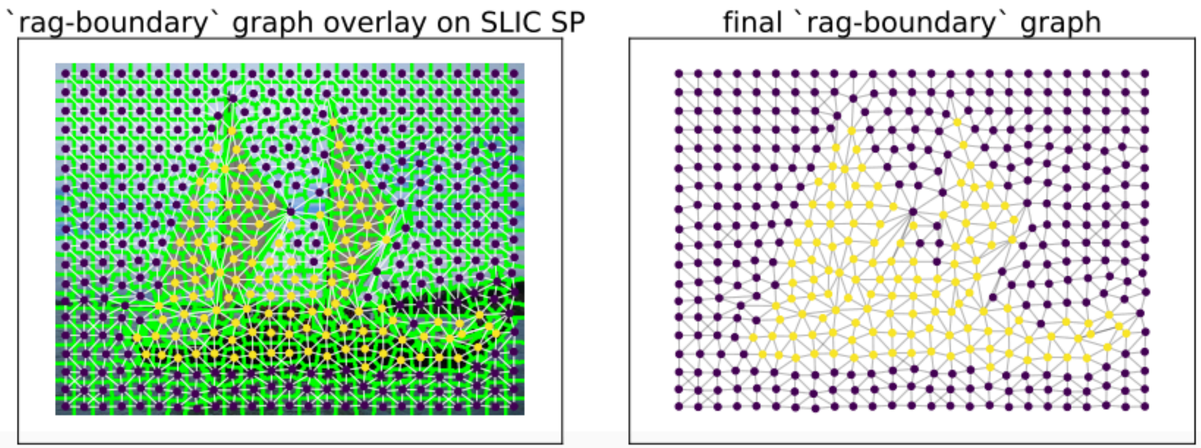

Table 2: Classification error on PascalVOC-SP.

Model	EGNN	AdS-GNN
Test F1 ↑	27.80 ± 0.74	28.07 ± 0.57